ZCAS University

CSC3631 WIRELESS AND MOBILE NETWORKS

FINAL EXAMINATION

1ST JUNE 2026

09:00 - 12:00 HOURS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. **Section A:** This section has only one question, which is compulsory and carries **40 Marks**
2. **Sections B**: Answer any **THREE** (3) questions from this section. Each question in this section carries 20 Marks
3. The total Marks for this exam are 100.
4. This exam paper consists of **FIVE** (5) Questions
5. Do not turn this page until you are told to do so.

SECTION A: COMPULSORY

QUESTION 1 – Case Study: ChizNet Wireless Mobility Failure

ChizNet Communications operates a nationwide wireless and mobile broadband infrastructure supporting universities, financial institutions, hospitals, and smart-city IoT deployments across Zambia. The company recently upgraded several metropolitan areas to support high-speed WiFi 6 (802.11ax), LTE, and 5G services. Following the upgrade, users reported persistent wireless communication failures. Investigations revealed severe signal attenuation in densely populated buildings, hidden-terminal problems in wireless LAN segments, excessive interference in the 2.4 GHz spectrum, and poor handover performance between LTE base stations. Some mobile users experienced dropped TCP sessions during mobility transitions, while overloaded access points caused congestion and retransmissions. Engineers further discovered improper channel allocation between neighbouring access points and weak mobility-management coordination between the Mobility Management Entity (MME), Serving Gateway (S-GW), and Home Subscriber Server (HSS). The incident disrupted online banking services, video conferencing systems, IoT telemetry transmissions, and cloud-based enterprise applications.

a) **a)** Identify three wireless-link characteristics that contributed to the communication failures and explain how each affected network performance. **(6 Marks)**

b) **b)** Differentiate between infrastructure mode and ad hoc mode wireless networking, giving one practical example of each from the scenario. **(4 Marks)**

c) **c)** Using the IEEE 802.11 standard, explain how CSMA/CA helps reduce collisions in wireless LANs and why collision detection is difficult in wireless environments. **(6 Marks)**

d) **d)** Discuss the causes and effects of the hidden terminal problem observed in ChizNet's wireless LAN deployment. **(4 Marks)**

e) Explain the role of the following LTE architecture components in mobility management:

i. Mobility Management Entity (MME)

ii. Serving Gateway (S-GW)

iii. Home Subscriber Server (HSS)

(6 Marks)

f) Describe how handover between LTE base stations occurs and explain why poor handover coordination may disrupt active user sessions. **(4 Marks)**

g) Compare 4G LTE and 5G networks in terms of transmission speed, latency, and deployment requirements. **(4 Marks)**

h) Explain the impact of wireless mobility on higher-layer protocols such as TCP and UDP. **(2 Marks)**

i) Recommend four measures ChizNet should implement to improve wireless security, reliability, and mobility performance across its infrastructure. **(4 Marks)**

[40 MARKS]

SECTION B: ANSWER ANY THREE QUESTIONS ONLY

QUESTION 2

a) Describe the major components of a wireless network and explain the role of each component. **(4 Marks)**

b) Explain how signal attenuation, interference, and multipath propagation affect wireless communication. **(6 Marks)**

c) Differentiate between wired and wireless networks in terms of transmission reliability and error handling. **(4 Marks)**

d) Discuss how SNR (Signal-to-Noise Ratio) influences wireless network performance and Bit Error Rate (BER). **(6 Marks)**

[20 MARKS]

QUESTION 3

a) Describe the advantages of representing cellular coverage areas using the hexagonal cell model. **(4 Marks)**

b) Compare macrocells, microcells, picocells, and femtocells in terms of coverage area, transmission power, and deployment environment. **(6 Marks)**

c) Discuss the major technical challenges faced in modern cellular networks and explain how technologies such as Massive MIMO, beamforming, and AI-driven optimisation help address them. **(4 Marks)**

d) Describe the process of handover (handoff) in cellular systems and explain the difference between hard handover and soft handover. **(6 Marks)**

[20 MARKS]

QUESTION 4

a) Explain the concepts of Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM) in wireless communication. **(4 Marks)**

b) Describe the operation of Bluetooth piconets and explain the roles of master and client devices. **(6 Marks)**

c) Discuss the advantages and limitations of 5G millimetre-wave communication. **(4 Marks)**

d) Explain how LTE sleep modes help conserve battery power in mobile devices. **(2 Marks)**

e) Discuss the importance of mobility management in supporting modern wireless and mobile applications. **(4 Marks)**

[20 MARKS]

QUESTION 5

a) Explain the role of the SIM card in GSM authentication and describe the importance of the following components stored within the SIM card:

i. IMSI
ii. Ki
iii. A3 algorithm
iv. A8 algorithm

(8 Marks)

b) Describe the purpose of ciphering in GSM communication and explain how it protects signalling and user data against interception. **(2 Marks)**

c) Discuss the role of the PIN code mechanism in GSM security and explain how it protects the subscriber when a SIM card is stolen or lost. **(4 Marks)**

d) Differentiate between IMSI and TMSI in GSM communication and explain how the use of TMSI improves user confidentiality and privacy. **(6 Marks)**

[20 MARKS]

END OF EXAMINATION